Vishnu Dutt Sharma

GitHub | Personal Webpage | LinkedIn

EDUCATION

University of Maryland, College Park *May 2020 - August 2024*
PhD in Computer Science
Advisor: Dr. Pratap Tokekar
Dissertation: Enhanced Robot Planning and Perception through Environment Prediction

Virginia Polytechnic Institute and State University *August 2019 - May 2020*
PhD Student, Computer Engineering
(*Transferred to University of Maryland, College Park*)

Indian Institute of Technology Kharagpur *July 2012 - June 2017*
Master of Technology, Bachelor of Technology in
Electronics & Electrical Communication Engineering
Specialization in Visual Information & Embedded Systems

PUBLICATIONS

- Hyojoon Park, Jeongran Lee, **Vishnu D. Sharma**, Aafiya Hussain, Atefeh Mohajeri Moghaddam, Zhiyi Wang, Matthew Andrews. Better Capture for Better Reconstruction: Understanding Capture Effects in 3D Gaussian Splatting. *Accepted at WACV 2027.*
- Harnaik Dhami, Charith Reddy, **Vishnu D. Sharma**, Troi Williams, Pratap Tokekar. GATSBI: An Online GTSP-Based Algorithm for Targeted Surface Bridge Inspection and Defect Detection. *IEEE TAES 2026.*
- **Vishnu D. Sharma***, Jeongran Lee*, Matthew Andrews, Ilija Hadžić. Hybrid Classical/RL Local Planner for Ground Robot Navigation. *RL Conference 2025.*
- Vishnu Sashank Dorbala*, **Vishnu D. Sharma***, Pratap Tokekar, Dinesh Manocha. Improving Zero-Shot ObjectNav with Generative Communication. *ICRA 2025.*
- Jeongran Lee, **Vishnu D. Sharma**, Lou Salaun, Matthew Andrews. Design of a Smooth Dynamic Digital Twin for Industrial Object Tracking. *IEEE MetaCom 2025.*
- Haiyuan Li, Hari Madhukumar, Nicholas Methley, Xuewen Chen, Yulei Wu, Juan Parra-Ullauri, **Vishnu D. Sharma**, Jeongran Lee, Arndt Ryo Koblitz, Matthew Andrews, Sige Liu, Yansha Deng, Oluwatayo Y. Kolawole, Andrea Tassi, Dimitra Simeonidou. Future Factories with 6G: Agentic AI and Cyber-Physical Digital Twins. *IEEE Internet of Things Journal, Oct 2025*
- Harnaik Dhami*, **Vishnu D. Sharma***, Pratap Tokekar. MAP-NBV: Multi-agent Prediction-guided Next-Best-View Planning for Active 3D Object Reconstruction. *IROS 2024.*
- Amisha Bhaskar, Rui Liu, **Vishnu D. Sharma**, Guangyao Shi, Pratap Tokekar. LAVA: Long-horizon Visual Action based Food Acquisition. *IROS 2024.*
- **Vishnu D. Sharma**, Anukriti Singh, Pratap Tokekar. Pre-Trained Masked Image Model for Mobile Robot Navigation. *ICRA 2024.*
- **Vishnu D. Sharma**, Jingxi Chen, Pratap Tokekar. ProxMaP: Proximal Occupancy Map Prediction for Efficient Indoor Robot Navigation. *IROS 2023.*
- Harnaik Dhami*, **Vishnu D. Sharma***, Pratap Tokekar. Pred-NBV: Prediction-guided Next-Best-View for 3D Object Reconstruction. *IROS 2023.*
- **Vishnu D. Sharma**, Lifeng Zhou, Pratap Tokekar. D2CoPlan: A Differentiable Decentralized Planner for Multi-Robot Coverage. *ICRA 2023.*

- Lifeng Zhou*, **Vishnu D. Sharma***, Qingbiao Li, Amanda Prorok, Alejandro Ribeiro, Pratap Tokekar, Vijay Kumar. Graph Neural Networks for Decentralized Multi-Robot Target Tracking. *IEEE SSRR 2022*.
- **Vishnu D. Sharma**, Maymoonah Toubeh, Lifeng Zhou, Pratap Tokekar. Risk-Aware Planning and Assignment for Ground Vehicles using Uncertain Perception from Aerial Vehicles. *IROS 2020*.
- Amrith Krishna, **Vishnu D. Sharma**, Bishal Santra, Aishik Chakraborty, Pavankumar Satuluri, Pawan Goyal. Poetry to Prose Conversion in Sanskrit as a Linearisation Task: A case for Low-Resource Languages. *ACL 2019*.
- Amrith Krishna, Bishal Santra, Sasi Prasanth Bandaru, Gaurav Sahu, **Vishnu D. Sharma**, Pavankumar Satuluri, Pawan Goyal. Free as in Free Word Order: An Energy Based Model for Word Segmentation and Morphological Tagging in Sanskrit. *EMNLP 2018*.
- Vikas Reddy, Amrith Krishna, **Vishnu D. Sharma**, Prateek Gupta, M. R. Vineeth, Pawan Goyal. Building a Word Segmenter for Sanskrit Overnight. *LREC 2018*.

Workshop Papers

- Junyu Mao, Sara Ayoubi, **Vishnu D. Sharma**, Ilija Hadžić, Matthew Andrews. SCOUT: Semantic scene COverage via Uncertainty-guided Traversal. (ICRA 2026 Workshop on Uncertainty in Open World Robotics).
- Anukriti Singh, **Vishnu D. Sharma**, Pratap Tokekar. FLIP-TD: Free Lunch Inpainting on Top-Down Images for Robotic Tasks. (ICRA 2023 Workshop on Pretraining for Robotics).
- **Vishnu D. Sharma**, Pratap Tokekar. Risk-Aware Path Planning for Ground Vehicles using Occluded Aerial Images. (ICRA 2022 Workshop on Robotic Perception and Mapping: Emerging Techniques).
- Shivani Nanda, **Vishnu D. Sharma**, Guangyao Shi, Jingxi Chen, Pratap Tokekar. Semantic Navigation for Assistive Robots: Identifying, Mapping, and Navigating to Door Signs in Indoor Environments. (IROS 2022 workshop on Social and Cognitive Interactions for Assistive Robotics Workshop).

**Equal Contribution*

Patent Application

- Daniel Andrews, Jeongran Lee, Vishnu Sharma. "Robot local planner selection". US20250093876A1, EP4524678B1, CN119620744A

WORK EXPERIENCE

Nokia of America Corporation (Nokia Bell Labs) Sep'24-Present

Robotics and Visual Computing Researcher, Modeling and Optimization Research Group

- Working on 2D and 3D map reconstruction and semantic scene understanding for industrial applications
- Exploring the intersection of robotics, 3D scene reconstruction, and foundational models

Nokia of America Corporation (Nokia Bell Labs) Jun'23-Aug'23

Robotics and 3D Modeling Intern, Modeling and Optimization Group

- Designed and implemented a hybrid local planner for ground robot navigation
- Showed the safe and efficient operation of the proposed robot through multiple experiments on a Clearpath Jackal robot
- Selected among the top 18 finalists across Bell Labs global internship program to present the work to the leadership team

Comcast Corporation May'21-Aug'21

Graduate Research Intern, Applied AI for CX Team

- Contributed to the improvement of the maintenance truck scheduling system using learning algorithms

- Designed a deep learning architecture to replace the heuristic-based variables, and weak-supervision rules for label improvement
- Implemented data monitoring pipelines for the existing data, and data processing modules for the new variables with PySpark

American Express India Pvt Ltd Jul'17 - Jul'19

Risk Analyst, Machine Learning & Data Science Team

- Developed deep learning models for applications including fraud detection, credit authorization and new accounts approval with specific focus on RNNs
- Contributed towards deep learning model deployment, benchmarking on cloud platform (AWS) and analyzing emerging techniques like distributed learning
- Implemented a variable creation tool using RNN aimed for enterprise-wide adoption

FlytBase Labs (Formerly *NavStik Labs*) May'16 - Jul'16

Summer Intern

- Integrated Semi-Direct Visual Odometry (SVO) package with FlytPOD flight computer for indoor localization
- Analyzed position estimation with visual odometry incorporating SVO for simple and complex trajectories in GPS-denied and GPS-assisted conditions
- Achieved indoor relocalization error contained within 10cm for linear and circular trajectories

American Express India Pvt Ltd May'15 - Jul'15

Summer Intern, Commerce Data Science Team

- Designed a categorization algorithm using SKU (Stock Keeping Unit) description for retail items
- Implemented a rule-based categorization system in Python for generating database by web-crawling and processing results for web search with Natural Language Processing Toolkit (NLTK) and other open-source libraries
- Achieved 60% accuracy in categorizing item across type, gender and brand dimensions

TECHNICAL STRENGTHS

Languages & Scripts	Python, C++, PySpark, Hive, MATLAB
Packages & Frameworks	PyTorch, TensorFlow, ROS, OpenCV, AirSim

TECHNICAL REPORTS AND PROJECTS

DeepVO: A Deep Learning approach for Monocular Visual Odometry
Implemented a CNN-based solution for Monocular Visual Odometry for ground robot (arXiv:1611.06069)

Interpretable Deep Reinforcement Learning for Green Security Games (GSG-I)
Developed an interpretable solution for Green Security Games with real-time information (arXiv:2211.04987)

Persistent-Monitoring using Multi-Robot (UAV-UGV) Coordination
Proposed a heuristic-based and a deep reinforcement learning-base approach for persistent monitoring by a team of UGVs and a UAV

Empirical Study of Second-Order Optimizers for Deep Learning Applications
Studied the effect of architecture and data on CNN and MLP with SGD, L-BFGS, and K-FAC optimizers for regression and classification tasks

Autonomous Aerial Vehicle
Worked on development of control stack of the Unmanned Aerial Vehicle. Part of the Best Team Cooperation Award-winning team at the International Aerial Robotics Competition 2016

Autonomous Underwater Vehicle
Contributed towards the development of the embedded and control stack of the underwater vehicle

Others Projects:
SmartKart (Semi-autonomous shopping cart), ROACH (Semi-autonomous all-terrain vehicle), Sirius (All-terrain vehicle with hybrid leg design), Plagiarism Detection in C++ source codes using NLP Tree Kernels

AWARDS AND ACHIEVEMENTS

- Kulkarni Foundation summer research fellowship (2022)
- Outstanding Student Research Award, Nokia Global Student Program (2023)
- Robotics and Automation Society travel grant for ICRA (2023)
- Jacob K. Goldhaber travel grant (2024)
- International Conference Student Support Award (2024)
- Member of the Best Team Cooperation Award-winning team at the International Aerial Robotics Competition (2016)
- Honorable mention at the IIT Kharagpur Gymkhana Awards for contributions to the technological activities (2017)
- All India Rank 933 in Indian Institute of Technology Joint Entrance Exam (IIT JEE)-2012 (amongst 0.5 million students)
- All India Rank 966 in All India Engineering Entrance Exam (AIEEE)-2012 (amongst 1.1 million students)

SERVICES

- Mentor at UPitchNJ 2026
- Volunteer at the 2022 Workshop on the Algorithmic Foundations of Robotics (WAFR)
- Mentor at AI4ALL summer camp at UMD College Park (2020, 2022)
- Conference and journal contributions reviewer
  - Conference on Robot Learning (CoRL): 2026
  - IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS): 2020, 2021, 2022, 2023, 2024, 2025
  - IEEE International Conference on Robotics and Automation (ICRA): 2023, 2024, 2025, 2026
  - International Conference on Automation Science and Engineering (CASE): 2023
  - International Symposium on Multi-Robot and Multi-Agent Systems (MRS): 2023
  - IEEE Transactions on Robotics (T-RO): 2022, 2023
  - IEEE Robotics and Automation Letters (RA-L): 2020, 2023, 2024, 2025

TEACHING

Teaching Assistant

- Decision-Making for Robotics (UMD, CMSC818B): Fall'23
- Introduction to Artificial Intelligence (UMD, CMSC421): Spring'21, Fall'21, Spring'22
- Applied Electrical Theory Lab (Virginia Tech, ECE2054): Spring'20
- Data Structures and Algorithms (Virginia Tech, ECE2574): Fall'19